

Q1 - 24 January - Shift 1

A travelling wave is described by the equation

$$y(x, t) = [0.05 \sin (8x - 4t)] \text{ m}$$

The velocity of the wave is : [all the quantities are in SI unit]

- (1) 4 ms^{-1} (2) 2 ms^{-1}
(3) 0.5 ms^{-1} (4) 8 ms^{-1}

Space for your notes:

Q2 - 25 January - Shift 1

The distance between two consecutive points with phase difference of 60° in a wave of frequency 500 Hz is 6.0 m. The velocity with which wave is traveling is _____ km/s

Space for your notes:

Q3 - 25 January - Shift 2

Match List I with List II

	List I		List II
A.	Troposphere	I.	Approximate 65-75 km over Earth's surface
B.	E-Part of Stratosphere	II.	Approximate 300 km over Earth's surface
C.	F ₂ -Part of Thermosphere	III.	Approximate 10 km over Earth's surface
D.	D-Part of Stratosphere	IV.	Approximate 100 km over Earth's surface

Choose the correct answer from the options given below :

- (1) A-III, B-IV, C-II, D-I
- (2) A-I, B-II, C-IV, D-III
- (3) A-I, B-IV, C-III, D-II
- (4) A-III, B-II, C-I, D-IV

Space for your notes:

Q4 - 25 January - Shift 2

A train blowing a whistle of frequency 320 Hz approaches an observer standing on the platform at a speed of 66 m/s. The frequency observed by the observer will be (given speed of sound = 330 ms^{-1}) Hz.

Space for your notes:

Q5 - 29 January - Shift 1

A person observes two moving trains, 'A' reaching the station and 'B' leaving the station with equal speed of 30 m/s. If both trains emit sounds with frequency 300 Hz, (Speed of sound : 330 m/s) approximate difference of frequencies heard by the person will be :

- (1) 33 Hz (2) 55 Hz
(3) 80 Hz (4) 10 Hz

Space for your notes:

Q6 - 29 January - Shift 1

Two simple harmonic waves having equal amplitudes of 8 cm and equal frequency of 10 Hz are moving along the same direction. The resultant amplitude is also 8 cm. The phase difference between the individual waves is _____ degree.

Space for your notes:

Q7 - 31 January - Shift 2

The displacement equations of two interfering waves are given by

$$y_1 = 10 \sin \left(\omega t + \frac{\pi}{3} \right) \text{ cm},$$
$$y_2 = 5 \left[\sin(\omega t) + \sqrt{3} \cos \omega t \right] \text{ cm} \text{ respectively. The}$$

amplitude of the resultant wave is _____ cm.

Space for your notes:

Q8 - 01 February - Shift 1

A steel wire with mass per unit length $7.0 \times 10^{-3} \text{ kg m}^{-1}$ is under tension of 70 N. The speed of transverse waves in the wire will be:

- (1) $200 \pi \text{ m/s}$ (2) 100 m/s
(3) 10 m/s (4) 50 m/s

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (3)**Q2 (18)****Q3 (1)****Q4 (400)****Q5 (2)****Q6 (120)****Q7 (20)****Q8 (2)**

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Q1 (3)

From the given equation $k = 8 \text{ m}^{-1}$ and $\omega = 4 \text{ rad/s}$

Velocity of wave = $\frac{\omega}{k}$

$$v = \frac{4}{8} = 0.5 \text{ m/s}$$

Q2 (18)

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta x$$

$$\frac{\pi}{3} = \frac{2\pi}{\lambda} (6\text{m})$$

$$\lambda = 36\text{m}$$

$$V = f\lambda = (500\text{Hz})(36\text{m})$$

$$= 18000 \text{ m/s} = 18 \text{ km/s}$$

Q3 (1)

NCERT fact based

Q4 (400)



$$f_{\text{app}} = f \left(\frac{v}{v - v_s} \right)$$

$$= 320 \left(\frac{330}{330 - 66} \right)$$

$$= 400 \text{ Hz}$$

Q5 (2)

$$f_1 = 300 \left(\frac{330 - 0}{330 - (-30)} \right) = 275$$

$$f_2 = 300 \left(\frac{330 - 0}{330 - (30)} \right) = 330$$

$$\Delta f = 330 - 275 = 55 \text{ Hz.}$$

Q6 (120)

$$2A \cos \left(\frac{\Delta \phi}{2} \right) = A$$

$$\cos \left(\frac{\Delta \phi}{2} \right) = \frac{1}{2}$$

$$\frac{\Delta \phi}{2} = 60^\circ$$

Q7 (20)

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No Solution

Q8 (2)

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{70}{70 \times 10^{-3}}} = 100 \text{ m/s}$$

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